Velocity and position graphs

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a position-time graph to velocity using slope.

SUCCESS CRITERIA - I can...

1. Read positions and elapsed times from a graph.
2. Find velocity from position change divided by time.
3. Explain the difference between speed and velocity.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Start at 3 m, move to 10 m, then finish at 5 m. Draw the route. Find distance, displacement with direction and final position.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $v_{avg} = \Delta x/\Delta t$; average speed = distance/Δt.
Δ means change; Δt is elapsed time. Position-time slope gives velocity.

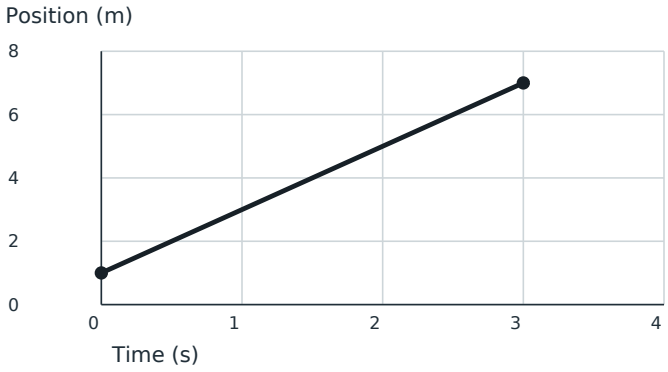
Worked reasoning: A cart moves from 1 m to 7 m in 3 s at steady velocity. The position-time slope is (7-1)/3=2 m/s.

Your turn: Sketch the graph through the two points. Draw and label the rise and run used for its slope.

Marzano check - circle: ① ② ③

03 Read the motion picture

The graph shows a cart moving right. Read its position at 0 s and 3 s. Draw an arrow showing its direction.



Marzano check - circle: ① ② ③

04 Find the rate

Use the graph in 3. Position change = ____ - ____ = ____ m. Time = ____ s. Velocity = ____ m/s to the ____.

Marzano check - circle: ① ② ③

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Continue down each column.

05 A pause

A position-time graph stays at 5 m from 4 s to 6 s. Is the cart moving? Explain what stays the same.

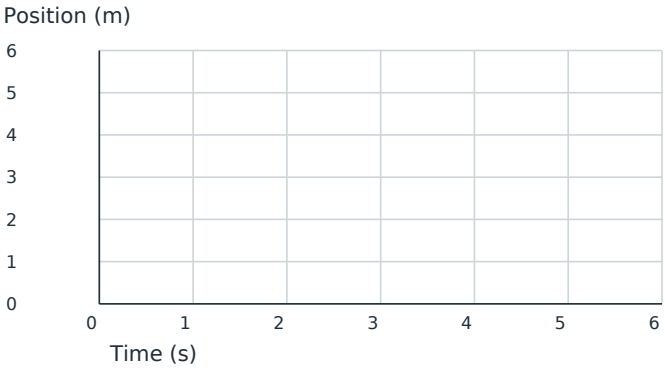
Support: The cart is ____ because its position ____.

Marzano check - circle: 1 2 3

06 Draw a walk with a pause

Plot the supplied points and connect neighbors with straight lines. Label moving right, stopped, and moving left. Use the blank graph.

Support: Data: t = 0, 2, 4, 6 s; x = 1, 5, 5, 1 m.



Marzano check - circle: 1 2 3

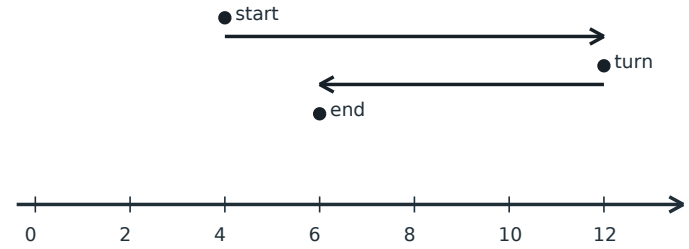
07 Distance or displacement?

For the walk in 6, find distance traveled and displacement. Why is average velocity zero even though the person walked?

Marzano check - circle: 1 2 3

08 Choose a method

Start at 4 m, move to 12 m, then finish at 6 m. Draw the route. Find distance, displacement with direction and final position.



Position (m); right is positive

Marzano check - circle: 1 2 3

09 Exit check

A cart goes from 9 m to 3 m in 5 s at steady velocity. Find the velocity with direction. Sketch the shape of its position-time graph.

Marzano check - circle: 1 2 3

10 STRETCH: Apply it another way

Draw two straight position-time graphs with different starting positions but the same velocity of 2 m/s. Explain the feature they share.

Marzano check - circle: 1 2 3